

Saksham Panghal

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EDUCATION

Indian Institute of Technology, Jodhpur

Bachelor of Technology , Computer Science & Engineering

CGPA: **8.55**

Aug. 2024 – May 2028

Central Board of Secondary Education

Senior Secondary Examination

Percentage: **93.6%**

Mar. 2024

Central Board of Secondary Education

Secondary School Examination

Percentage: **99.0%**

Mar. 2022

EXPERIENCE

Undergraduate Researcher at SPADE Lab

IIT Jodhpur

Jan. 2026 – Apr. 2026

Jodhpur, Rajasthan

- Worked on the problem of **component classification** in complex graph-structured datasets where varying **graph topology, sparsity, and class imbalance** reduce the effectiveness of static learning pipelines
- Addressed severe **class imbalance** using techniques such as **focal loss** and **adaptive sampling**
- Built an **automated benchmarking pipeline** to evaluate model based on accuracy, macro-F1, and balanced accuracy on Dense, Sparse and Imbalanced graph datasets
- Improved the maximum **f1-score** on **imbalanced real world graph** datasets to **92.1%**, an increase of **22.7%** from traditional approaches.

PROJECTS

Collaborative Code Workspace | Node.js, Yjs, PostgreSQL, Socket.io, Docker |

Apr. 2026 – Jun. 2026

- Built a **real-time collaborative coding workspace** using React, Node.js, **Socket.IO**, and **Yjs**, supporting multi-user editing, file management, and **low-latency CRDT**-based synchronization.
- Designed a hybrid persistence architecture with active workspace state maintained in memory and **automatic PostgreSQL persistence**, reducing database overhead while enabling reliable workspace recovery.
- Engineered room lifecycle management, user presence tracking, and autosave workflows to support scalable and resilient collaborative sessions.

College Utility Platform | Flutter, Firebase, FastAPI, Python, Dart |

Feb. 2026 – Apr. 2026

- Developed a cross-platform mobile application using **Flutter** for campus utility services including **lost & found and ride sharing**
- Implemented **real-time updates** for ride matching and item claim status using Firebase listeners
- Structured the app with **modular architecture** and efficient state management for scalability and maintainability

Metro Route Planner | Node.js, React, PostgreSQL, Redis, Docker |

Mar. 2026 - May 2026

- Developed a **full-stack metro navigation system** for computing optimal routes, estimated travel time, geographical distance and fare between stations using **graph based** and other algorithms
- Implemented **shortest time** and **minimum stations** routing using **Dijkstra's Algorithm and BFS** after generating a weighted metro network graph from **GTFS** data
- Optimized query performance and reduced redundant computations through graph preprocessing and caching strategies, decreasing query latency by **4500 ms** per request

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: React, Node.js, Next.js, FastAPI, Pandas, Matplotlib

Developer Tools: Git, Docker, VS Code, Visual Studio, Postman

Database: PostgreSQL, MySQL, MongoDB, Firebase, Redis

RELEVANT COURSEWORK

Data Structures and Algorithms · Mathematics for Computing · Software Engineering · Pattern Recognition and ML ·

Introduction to Computer Science · Probability, Statistics & Stochastic Processes

ACHIEVEMENTS

- Achieved an All India Rank of **2480** in JEE Advanced 2024
- Achieved **99.81 percentile** in JEE Main 2024, securing a place in **top 0.19%** of **1.4M+** students from all over India.